BifocalGemma: Asymmetric Bidirectional Repository Encoding and Stationary Cross-Attention for Gemma 4 Developer Agents

Buana Daru Okta
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`https://github.com/daruokta/bifocalgemma`
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Abstract

Autoregressive decoder-only language models process software repositories under a fundamental structural bottleneck: causal attention masks prevent earlier repository tokens from attending to symbols, type definitions, or call sites that appear later in the linearized file stream, while prefix-concatenated prompts force the decoder to maintain massive key-value (KV) caches across tens of thousands of tokens during patch generation. We introduce **BifocalGemma**, an asymmetric encoder-decoder developer agent architecture built natively on the **Gemma 4** model family (`Gemma 4 E2B` and `Gemma 4 E4B`) [4]. BifocalGemma decouples repository comprehension from autoregressive patch synthesis by pairing a bidirectional Gemma 4 repository encoder ($N_{\text{enc}} \leq$ 32,768 tokens) with a causal Gemma 4 patch decoder ($N_{\text{dec}} \leq$ 8,192 tokens). To bridge these streams without destabilizing pretrained causal representations or breaking Gemma 4's native cross-layer KV-sharing topology, we introduce three architectural innovations: (1) a **SwiGLU Dual-Gate Coordinate Bridge** (`SwiGLUDualGateBridge`) with orthogonal/identity residual pathways, zero-initialized residual entry, and per-channel post-normalization calibration; (2) **Decoupled Stationary Full-Rank Cross-Attention** (`BifocalAttentionBlock`) that enforces rotational position invariance ($\Theta_{\text{cross}} = 0$), logarithmic **Critical Attention Length Scaling** $\beta(N_{\text{enc}})$, learnable per-head sharpness scaling σ_{cross}, and dual-gated head- and token-level fusion; and (3) a **Two-Stage Paired Training Curriculum** driven by **Chunked Differential Diff Cross-Entropy** ($\mathcal{L}_{\text{diff}}$). Across the 6-model controlled ablation matrix evaluated on `LongCodeArena` (`SWE-QA`) [2], `RepoBench-C-8k` [7], and `SWE-bench Lite` [6] under a strict Neutral Prompt Contract, **[PLACEHOLDER: Quantitative summary of primary benchmark outcomes across SWE-QA, RepoBench-C-8k, and SWE-bench Lite, alongside 32K memory and decoding throughput gains, to be populated upon completion of the active runs]**. All models are exported as standalone Hugging Face `PreTrainedModel` artifacts (`BifocalGemmaForCausalLM`) supporting native `AutoModelForCausalLM.from_pretrained(..., trust_remote_code=True)`.

1 Introduction

Repository-level software engineering requires two distinct computational modes: *global bidirectional comprehension* over multi-file dependency graphs and *precise local autoregressive synthesis* of unified diffs. Standard causal language models conflate both modes into a single left-to-right token stream. When a multi-file repository snapshot is linearized into a causal prefix $x_1, \ldots, x_N$, a declaration at token x_i cannot attend to downstream callers or interface implementations at token x_j $(j > i)$. Furthermore, during iterative reasoning and patch generation, every generated decoder token must attend over the entire N-token causal KV cache, incurring $\mathcal{O}(N_{\text{enc}}+N_{\text{dec}})$ memory bandwidth overhead at every decoding step.

Recent efforts to adapt causal backbones into bidirectional encoders [1, 3] demonstrate that unmasking self-attention unlocks strong bidirectional representations. However, grafting a bidirectional repository encoder onto a modern **Gemma 4** [4] decoder presents three acute architectural challenges:

1. **Subspace and Norm Mismatch Under Unmasked Self-Attention**: Removing the causal mask from a pretrained Gemma 4 backbone alters the spectral distribution and activation norms of the final hidden states, causing variance explosion if injected directly into a causal decoder.

2. **RoPE Coordinate Collision and Long-Context Entropy Dilution**: Standard Rotary Position Embeddings (RoPE) [9] assign sequential positions $t \in [0, N-1]$. Applying RoPE across decoupled encoder

and decoder streams creates artificial distance decay between a decoder token at step t_{dec} and relevant repository definitions at arbitrary file offsets $t_{\text{enc}} \gg t_{\text{dec}}$. Meanwhile, attending over $N_{\text{enc}} \in [8\text{K}, 32\text{K}]$ encoder tokens increases softmax entropy as $\mathcal{O}(\ln N_{\text{enc}})$, washing out sharp symbol lookups.

3. **Catastrophic Forgetting During Adapter Warmup**: Jointly updating decoder weights alongside randomly initialized cross-attention projections corrupts the decoder's fluent code generation capabilities before the cross-attention bridge has aligned its coordinate system.

Contributions. We present **BifocalGemma**, a modular encoder-decoder architecture engineered specifically for Gemma 4 developer agents:

- **Asymmetric Bidirectional-to-Causal Gemma 4 Topology**: We pair unmasked bidirectional Gemma 4 encoders (`E2B` at $d = 1536$, 35 layers; `E4B` at $d = 2560$, 42 layers) with causal Gemma 4 decoders while strictly isolating Per-Layer Embeddings (PLE, $d_{\text{ple}} = 256$) per stream and preserving Gemma 4's native cross-layer KV-sharing hierarchy (15 KV-owner + 20 KV-shared layers in `E2B`; 24 KV-owner + 18 KV-shared layers in `E4B`).

- **SwiGLU Dual-Gate Coordinate Bridge**: We design a residual manifold projector with width $d_{\text{mid}} = \max(d_{\text{enc}}, d_{\text{dec}})$, identity/orthogonal shortcut connections, zero-initialized down-projection, and learnable per-channel calibration γ_{calib} that seamlessly supports both symmetric (`E2B`→`E2B`, `E4B`→`E4B`) and heterogeneous (`E2B`→`E4B`, `E4B`→`E2B`) pairings.

- **Stationary Full-Rank Cross-Attention with Critical Length Scaling**: We formulate a RoPE-free ($\Theta_{\text{cross}} = 0$) cross-attention mechanism equipped with logarithmic length scaling $\beta(N_{\text{enc}})$, learnable per-head sharpness $\sigma_{\text{cross}} \in [0.1, 5.0]$, and a dual-gated injection mechanism initialized at 5% amplitude ($\tanh(g^{(0)}_{\text{head}}) = 0.05$).

- **Two-Stage Paired Curriculum & Differential Diff Objective**: We combine a frozen-decoder bridge alignment stage (Stage 1) and a conservative joint LoRA [5] SFT stage with 20% comprehension replay (Stage 2) under a memory-efficient Chunked Differential Diff Cross-Entropy loss ($\mathcal{L}_{\text{diff}}$).

- **Standalone Hugging Face Ecosystem Integration**: We package BifocalGemma as a self-contained Hugging Face `PreTrainedModel` (`BifocalGemmaForCausalLM`) with bundled architecture code and self-healing configuration assets.

2 Architecture and Methodology

2.1 Asymmetric Gemma 4 Backbone and Bidirectional Repository Encoder

Given a repository context X_{enc} of length $N_{\text{enc}} \leq 32{,}768$ and a task instruction q, we prepend a lightweight query prefix (`[QUERY] <instruction>\n`) to X_{enc} so that encoder self-attention layers condition repository representations on the target issue from layer 0. The encoder backbone is initialized from `Gemma 4` (`E2B` with $d_{\text{enc}} = 1536$, $L = 35$ layers, $H_Q = 8$, $H_{KV} = 1$; or `E4B` with $d_{\text{enc}} = 2560$, $L = 42$ layers, $H_Q = 8$, $H_{KV} = 2$). As visualized in Figure 1 (Left), the encoder processes the entire multi-file repository context in a single global bidirectional pass.

To convert the causal Gemma 4 backbone into a true bidirectional repository encoder:

1. **Unmasked Bidirectional Attention**: We override the causal attention mask across all sliding-window and global full-attention layers (`use_bidirectional_attention="all"`), constructing a non-causal 4D padding mask $M_{\text{enc}} \in \{0, -\infty\}^{B\times 1\times 1\times N_{\text{enc}}}$ that permits every valid repository token i to attend to both preceding ($j < i$) and succeeding ($j > i$) tokens while strictly masking padding positions.

2. **Strict Stream-Isolated PLE**: Gemma 4 utilizes Per-Layer Embeddings (`hidden_size_per_layer_input` = 256) tied to surface token identities. We compute $E^{\text{enc}}_{\text{ple}}$ strictly from X_{enc} and $E^{\text{dec}}_{\text{ple}}$ strictly from X_{dec}, preventing cross-stream token identity leakage.

3. **Post-Encoding Padding Zeroing**: Output hidden states at padded positions are explicitly zeroed ($H_{\text{enc}} \leftarrow H_{\text{enc}} \odot \mathbb{I}[\text{mask}_{\text{enc}} = 1]$) before entering the coordinate bridge.

2.2 SwiGLU Dual-Gate Coordinate Bridge

To project bidirectional encoder representations $H_{\text{enc}} \in \mathbb{R}^{B\times N_{\text{enc}}\times d_{\text{enc}}}$ into the causal decoder's target manifold $Z_{\text{enc}} \in \mathbb{R}^{B\times N_{\text{enc}}\times d_{\text{dec}}}$ across arbitrary encoder-decoder width pairings, `SwiGLUDualGateBridge` applies a pre-normalized gated residual transformation [10, 8] with intermediate dimension $d_{\text{mid}} = \max(d_{\text{enc}}, d_{\text{dec}})$:

$$\tilde{H} = \text{RMSNorm}_{\text{pre}}(H_{\text{enc}}), \tag{1}$$

$$G = \text{SiLU}(\tilde{H}W_{\text{gate}}) \odot (\tilde{H}W_{\text{up}}), \quad W_{\text{gate}}, W_{\text{up}} \in \mathbb{R}^{d_{\text{enc}}\times d_{\text{mid}}}, \tag{2}$$

$$Z = W_{\text{res}}(\tilde{H}) + GW_{\text{down}}, \quad W_{\text{down}} \in \mathbb{R}^{d_{\text{mid}}\times d_{\text{dec}}}, \tag{3}$$

$$Z_{\text{enc}} = (\gamma_{\text{calib}} \odot \text{RMSNorm}_{\text{post}}(Z)) \odot \mathbb{I}[\text{mask}_{\text{enc}} = 1]. \tag{4}$$

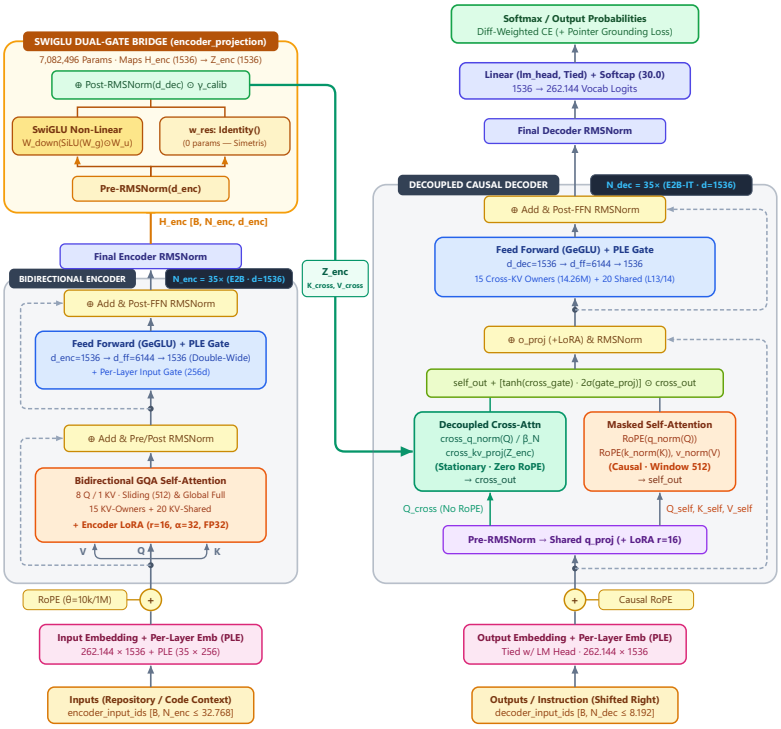


Figure 1: **The BifocalGemma V3 Model Architecture.** Comprehensive bottom-to-top structural dataflow following the classic Transformer paradigm: (**Left**) The Bidirectional Repository Encoder Stack ($N_{\text{enc}} \leq 32{,}768$) with stream-isolated Per-Layer Embeddings (PLE, $d_{\text{ple}} = 256$), unmasked self-attention across sliding and global heads, and final encoder RMSNorm; (**Center-Top**) The SwiGLU Dual-Gate Coordinate Bridge (SwiGLUDualGateBridge) with intermediate expansion $d_{\text{mid}} = \max(d_{\text{enc}}, d_{\text{dec}})$, identity/orthogonal residual pathway W_{res}, and per-channel scale calibration γ_{calib}; (**Center**) The Z_{enc} corridor highway conveying stationary Key/Value representations to decoder cross-attention; and (**Right**) The Decoupled Causal Decoder Stack ($N_{\text{dec}} \leq 8{,}192$) with parallel decoupled cross-attention ($\Theta_{\text{cross}} = 0$) and causal self-attention, dual-gated head and token fusion, tied language model head, and logit softcapping.

Here, W_{res} is an identity mapping when $d_{\text{enc}} = d_{\text{dec}}$ (`E2B`→`E2B`, `E4B`→`E4B`) and an orthogonally initialized linear projection (3,932,160 parameters) when $d_{\text{enc}} \neq d_{\text{dec}}$ (`E2B`→`E4B`, `E4B`→`E2B`). The down-projection W_{down} is initialized near zero ($\mathcal{N}(0, (0.01/\sqrt{d_{\text{mid}}})^2)$) so the bridge begins as a clean linear/identity manifold transport, while $\gamma_{\text{calib}} \in \mathbb{R}^{d_{\text{dec}}}$ (initialized to $\mathbf{1}$) provides learnable per-channel scale calibration prior to cross-attention key/value projection.

Table 1 provides the exact parameter budgets across all components for the complete 6-model matrix. In symmetric pairings (`Bifocal-2B` and `Bifocal-4B`), the bridge introduces only 7.08M and 19.67M parameters ($< 0.15\%$ of model scale), while asymmetric projections scale smoothly to 18.36M and 20.98M. During Stage 2 joint SFT, LoRA adapter parameterization ($r = 16, \alpha = 32$) restricts trainable updates to 32.06M (0.36%) in `Bifocal-2B` and 111.34M (0.78%) in `Bifocal-4B`, maintaining exceptional parameter efficiency.

2.3 Decoupled Stationary Full-Rank Cross-Attention

Every decoder layer wraps the native Gemma 4 attention module inside a `BifocalAttentionBlock` that executes causal self-attention and encoder-decoder cross-attention in parallel before merging their outputs through the shared output projection W_O (Figure 1, Right).

Preservation of Gemma 4 Cross-Layer KV-Sharing. Gemma 4 employs a hybrid KV-sharing architecture in which only the first 15 of 35 layers in `E2B` (and 24 of 42 layers in `E4B`) compute independent KV projections (`is_kv_shared_layer = False`), while downstream shared layers reuse the KV states of their designated owner layer (Figure 2 and Table 2). BifocalGemma mirrors this exact invariant for cross-attention: only KV-owner layers instantiate dedicated BF16 full-rank projections $W_{K,\text{cross}}, W_{V,\text{cross}} \in \mathbb{R}^{d_{\text{dec}} \times (H_{KV} \cdot d_h)}$ and RMSNorms ($\text{RMSNorm}_{K,\text{cross}}, \text{RMSNorm}_{V,\text{cross}}$), warm-started from the pretrained self-attention W_K, W_V weights. During autoregressive generation, $(K_{\text{cross}}, V_{\text{cross}})$ are computed once

Table 1: Comprehensive Parameter Budgets and Allocation across the 6-Model Controlled Matrix. Bridge parameters include SwiGLU projections ($W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}$), residual routing W_{res} (identity for symmetric; 3.93M orthogonal linear for asymmetric), norms, and per-channel calibration γ_{calib}. Cross-attention parameters scale strictly with KV-owner layers (15 in `E2B`; 24 in `E4B`). Stage 2 joint training activates LoRA ($r = 16, \alpha = 32$) on query/key/value/output linear weights, updating $< 0.9\%$ of total model parameters.

Model ID	Hidden Dimensions ($d_{\text{enc}} \to d_{\text{mid}} \to d_{\text{dec}}$)	Enc / Dec Layers	Bridge Params	Cross-Attn Params	LoRA ($r = 16$) (Enc + Dec)	Total Params (Tied)	Total Params (Untied)	Stage 1 Trainable (Bridge + Cross)	Stage 2 Trainable (Joint SFT)
`Baseline-E2B-IT`	$0 \to 0 \to 1536$	0 / 35L	—	—	0.00M + 5.36M	4.63B	4.63B	—	5.36M (0.12%)
`Baseline-E4B-IT`	$0 \to 0 \to 2560$	0 / 42L	—	—	0.00M + 9.08M	7.46B	7.46B	—	9.08M (0.12%)
`Bifocal-2B`	$1536 \to 1536 \to 1536$	35L / 35L	7.08M	14.26M	5.36M + 5.36M	8.88B	9.28B	21.34M (0.24%)	32.06M (0.36%)
`Bifocal-4B-E2B`	$1536 \to 2560 \to 2560$	35L / 42L	18.36M	73.52M	5.36M + 9.08M	N/A	12.18B	91.88M (0.75%)	106.31M (0.87%)
`Bifocal-2B-E4B`	$2560 \to 2560 \to 1536$	42L / 35L	20.98M	14.26M	9.08M + 5.36M	N/A	12.13B	35.23M (0.29%)	49.67M (0.41%)
`Bifocal-4B`	$2560 \to 2560 \to 2560$	42L / 42L	19.67M	73.52M	9.08M + 9.08M	14.35B	15.02B	93.19M (0.65%)	111.34M (0.78%)

Table 2: Architectural Specifications and Cross-Layer KV-Sharing Topology for the `Gemma 4` Backbone Stacks. In both encoder and decoder, only KV-owner layers instantiate independent key/value projections. For cross-attention, downstream KV-shared layers completely bypass Cross-KV projection, reusing cached states from designated Sliding and Global Anchor layers with $\mathcal{O}(1)$ step overhead.

Structural Attribute	`Gemma-4-E2B`	`Gemma-4-E4B`
Hidden Dimension (d)	1536	2560
Total Transformer Layers (L)	35	42
Query Heads (H_Q)	8 ($d_h = 256/512$)	8 ($d_h = 256/512$)
Key-Value Heads (H_{KV})	1 (GQA 8:1)	2 (GQA 4:1)
Per-Layer Embedding (d_{ple})	256 (Stream-Isolated)	256 (Stream-Isolated)
Global Full-Attention Layers	7 / 35 (20.0%)	7 / 42 (16.67%)
Layer Indices	`L04, 09, 14, 19, 24, 29, 34`	`L05, 11, 17, 23, 29, 35, 41`
Global Head Dimension	512 (25% RoPE Rotated)	512 (25% RoPE Rotated)
Sliding Window Layers	28 / 35 (80.0%)	35 / 42 (83.33%)
Window Size / Head Dim	$w = 512$ / $d_h = 256$	$w = 512$ / $d_h = 256$
KV-Owner Layers	15 / 35 (42.86%)	24 / 42 (57.14%)
Dedicated Cross-KV Proj.	`L00` – `L14`	`L00` – `L23`
KV-Shared Layers	20 / 35 (57.14%)	18 / 42 (42.86%)
Reused Anchor Layers	`L15` – `L34`	`L24` – `L41`
Designated Anchor Layers		
Sliding Window Anchor	`L13` (`store_full_kv`)	`L22` (`store_full_kv`)
Global Full Anchor	`L14` (`store_full_kv`)	`L23` (`store_full_kv`)
Cross-KV Memory Reduction	**57.14%**	**42.86%**

during prefill, cached inside `BifocalDynamicCache`, and reused across both KV-shared layers and subsequent decoding tokens with $\mathcal{O}(1)$ recomputation cost, yielding a 57.14% memory reduction on `E2B` and 42.86% on `E4B`.

Stationary Coordinate Invariance ($\Theta_{\text{cross}} = 0$). While causal self-attention applies standard RoPE rotations $\mathcal{R}_{\Theta,t}$ to Q_{self} and K_{self}, applying RoPE to cross-attention introduces spurious positional interference between the decoder's generation index and the encoder's file offset. Because Z_{enc} already encodes intra-file and inter-file structural topology from 35–42 layers of RoPE-equipped bidirectional self-attention, we enforce *Stationary Coordinate Invariance* ($\Theta_{\text{cross}} = 0$) on the cross-attention branch:

$$Q_{\text{cross}} = \text{RMSNorm}_{Q,\text{cross}}(H_{\text{dec}} W_Q), \tag{5}$$

$$K_{\text{cross}} = \text{RMSNorm}_{K,\text{cross}}(Z_{\text{enc}} W_{K,\text{cross}}), \tag{6}$$

$$V_{\text{cross}} = \text{RMSNorm}_{V,\text{cross}}(Z_{\text{enc}} W_{V,\text{cross}}). \tag{7}$$

Critical Attention Length Scaling and Learnable Sharpness. To prevent attention entropy explosion when querying up to $N_{\text{enc}} = 32{,}768$ repository tokens, we modulate Q_{cross} with both a deterministic logarithmic length normalizer $\beta(N_{\text{enc}})$ and a learnable per-head sharpness parameter $\sigma_{\text{cross}} \in \mathbb{R}^{H_Q \times 1 \times 1}$ (initialized to 1.0 and clamped to $[0.1, 5.0]$):

$$\beta(N_{\text{enc}}) = \max\left(1.0,\ 1.0 + 0.1 \ln\left(\frac{\max(N_{\text{enc}}, 1)}{2048}\right)\right), \tag{8}$$

$$\hat{Q}_{\text{cross}} = \frac{\text{clamp}(\sigma_{\text{cross}}, 0.1, 5.0)}{\beta(N_{\text{enc}})} Q_{\text{cross}}, \tag{9}$$

$$O_{\text{cross}} = \text{SDPA}\left(\hat{Q}_{\text{cross}}, K_{\text{cross}}, V_{\text{cross}}, M_{\text{enc}}\right). \tag{10}$$

Dual-Gated Head and Token Fusion. Rather than summing O_{self} and O_{cross} unconditionally, we combine a global per-head gate $g_{\text{head}} \in \mathbb{R}^{1 \times H_Q \times 1 \times 1}$ (initialized to $\text{atanh}(0.05)$ so $\tanh(g^{(0)}_{\text{head}}) = 0.05$) with a fine-grained token-conditional query gate parameterized by $w_{\text{gate}} \in \mathbb{R}^{1 \times H_Q \times 1 \times d_h}$ (initialized to $\mathbf{0}$ so $g^{(0)}_{\text{query}} = 1.0$):

$$g_{\text{query}} = 2\,\sigma\left(\frac{4}{\sqrt{d_h}} \langle Q_{\text{cross}}, w_{\text{gate}} \rangle\right) \in (0, 2), \tag{11}$$

$$g_{\text{eff}} = \tanh(g_{\text{head}}) \cdot g_{\text{query}}, \tag{12}$$

$$O_{\text{total}} = W_O\,(O_{\text{self}} + g_{\text{eff}} \odot O_{\text{cross}}). \tag{13}$$

At step 0 of training, cross-attention contributes a bounded 5% perturbation to the pre-projection attention representation, protecting the decoder's causal prior while preserving non-zero gradients into both the bridge and cross-attention projections.

2.4 Chunked Differential Diff Objective and Two-Stage Curriculum

Chunked Differential Diff Cross-Entropy ($\mathcal{L}_{\text{diff}}$). Standard uniform token-level cross-entropy underweights critical edit operations (`+` additions and `-` deletions) relative to boilerplate context lines inside unified diffs. We optimize *Chunked Differential Diff Cross-Entropy* ($\mathcal{L}_{\text{diff}}$), which parses target patches into line-level syntactic categories and assigns differential token weights: $w_{\text{add}} = 1.4$ for added lines (`+`), $w_{\text{del}} = 1.2$ for deleted lines (`-`), $w_{\text{hdr}} = 1.1$ for hunk headers (`@@ ... @@`, `---`, `+++`), and $w_{\text{ctx}} = 0.8$ for unchanged context

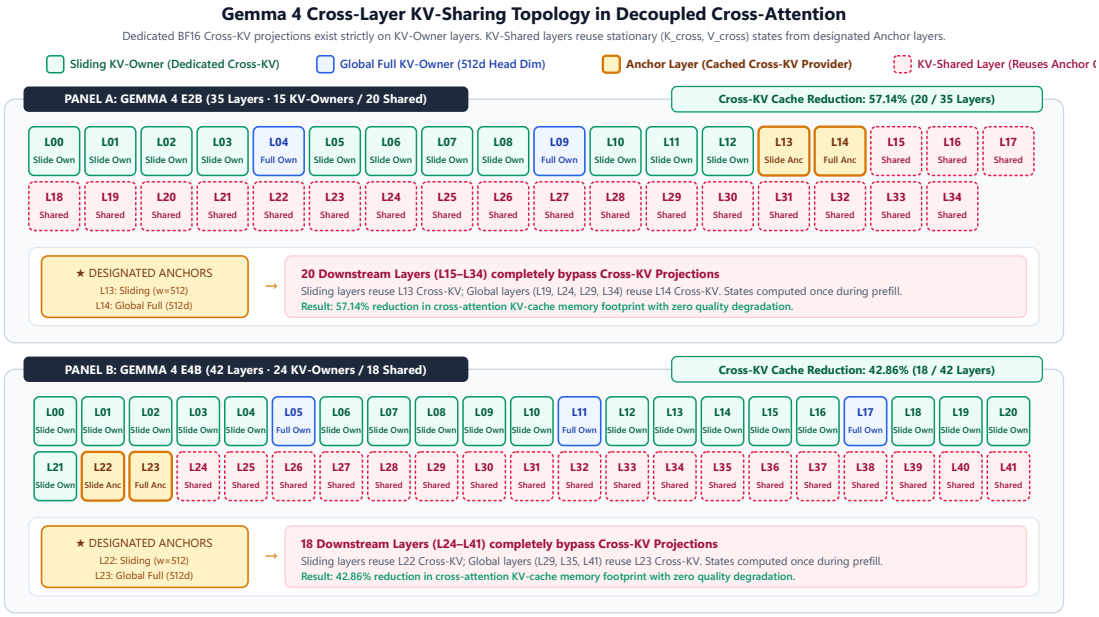


Figure 2: **Gemma 4 Cross-Layer KV-Sharing Topology in Decoupled Cross-Attention.** Structural layer allocation across the 35 layers of `Gemma-4-E2B` (Panel A) and 42 layers of `Gemma-4-E4B` (Panel B). Dedicated BF16 Cross-KV projections are instantiated exclusively on KV-owner layers (15 in `E2B`; 24 in `E4B`). Subsequent KV-shared layers (20 in `E2B`, 57.14%; 18 in `E4B`, 42.86%) reuse stationary $(K_{\text{cross}}, V_{\text{cross}})$ states computed by designated Anchor layers (`L13`/`L14` in `E2B`; `L22`/`L23` in `E4B`), reducing cross-attention KV-cache storage by up to 57.14% with $\mathcal{O}(1)$ step overhead.

lines (Table 3). Weights are normalized per sequence to maintain unit mean ($\frac{1}{|S|}\sum_{t\in S}\tilde{w}_t = 1.0$). To minimize peak VRAM on $d_{\text{vocab}} = 262{,}144$, logits are computed in chunks of 128 tokens with Gemma 4 final logit softcapping (30.0):

$$\mathcal{L}_{\text{diff}} = \frac{1}{|S|}\sum_{t\in S}\tilde{w}_t\,\text{CE}\left(30.0\tanh\left(\frac{h_t W_{\text{lm}}^{\top}}{30.0}\right), y_t\right). \quad (14)$$

Streaming pointer-generator copy loss is explicitly disabled ($\lambda_{\text{ptr}} = 0.0$) to prevent verbatim copy bias when synthesizing novel bug-fix tokens.

Two-Stage Paired Curriculum vs. Fair Baseline Curriculum.

- **Stage 1 (Frozen Decoder Bridge Alignment — First 50% of Steps)**: Both the encoder and decoder backbones are strictly frozen. Only `SwiGLUDualGateBridge` and the KV-owner `BifocalAttentionBlock` parameters are trained at $\text{LR}_{\text{bridge}} = \text{LR}_{\text{cross}} = 10^{-4}$ using comprehension and localization tasks.
- **Stage 2 (Joint LoRA SFT with Anti-Forgetting Replay — Final 50% of Steps)**: LoRA adapters ($r = 16, \alpha = 32$) are activated on both encoder and decoder with a conservative decoder learning rate $\text{LR}_{\text{lora}} = 2\times10^{-5}$, while bridge and cross-attention learning rates are halved to 5×10^{-5}. Stage 2 interleaves 80% unified diff synthesis (`SWE-smith`) with a 20% comprehension replay buffer (`SWE-QA` / `RepoBench`) to preserve long-

Table 3: Operational Hyperparameters, Gating Initializations, and Optimization Schedule for BIFOCALGEMMA. All gating and scaling factors are formulated to preserve causal fluency at initialization while permitting stable cross-stream gradient flow.

Component / Hyperparameter	Operational Setting / Formula
Decoupled Cross-Attention Gating & Scaling	
Cross-Attention RoPE Frequency	$\Theta_{\text{cross}} = 0$ (Stationary Coordinate Invariant)
Global Head Gate g_{head} Init	$\text{atanh}(0.05) \implies \tanh(g_{\text{head}}^{(0)}) = 0.05$
Query Token Gate w_{gate} Init	$\mathbf{0} \implies g_{\text{query}}^{(0)} = 1.0 \implies g_{\text{eff}}^{(0)} = 0.05$
Attention Length Scaling $\beta(N_{\text{enc}})$	$\max(1.0,\ 1.0 + 0.1\ln(\max(N_{\text{enc}}, 1)/2048))$
Per-Head Sharpness σ_{cross}	$\sigma_{\text{cross}}^{(0)} = 1.0$, clamped to $[0.1, 5.0]$
SwiGLU Dual-Gate Coordinate Bridge	
Intermediate Width d_{mid}	$\max(d_{\text{enc}}, d_{\text{dec}})$ (1536 or 2560)
Down-Projection W_{down} Init	$\mathcal{N}(0, (0.01/\sqrt{d_{\text{mid}}})^2)$ (Zero-Residual Entry)
Residual Connection W_{res}	Identity ($d_{\text{enc}} = d_{\text{dec}}$); Orthogonal Linear ($d_{\text{enc}} \neq d_{\text{dec}}$)
Channel Calibration γ_{calib} Init	$\mathbf{1} \in \mathbb{R}^{d_{\text{dec}}}$ (Unit Scale Prior)
Chunked Differential Diff Objective ($\mathcal{L}_{\text{diff}}$)	
Added Lines Token Weight (w_{add})	1.4 (Diff Line Prefix +)
Deleted Lines Token Weight (w_{del})	1.2 (Diff Line Prefix -)
Hunk Headers Token Weight (w_{hdr})	1.1 (Diff Line Prefix `@@ ... @@`)
Context Lines Token Weight (w_{ctx})	0.8 (Unmodified Context Lines)
Logit Softcapping Threshold	$30.0 \cdot \tanh(\text{logits}/30.0)$
Vocabulary Logit Chunk Size	128 tokens ($d_{\text{vocab}} = 262{,}144$)
Differentiable Pointer Weight λ_{ptr}	0.0 (Disabled to prevent copy bias)
Two-Stage Paired Curriculum Schedule	
Maximum Context Allocation	$N_{\text{enc}} \leq 32{,}768$ (80%) : $N_{\text{dec}} \leq 8{,}192$ (20%)
Stage 1 Duration	First 50% of Total Steps (Frozen Decoder)
Learning Rates (Stage 1)	$\text{LR}_{\text{bridge}} = \text{LR}_{\text{cross}} = 10^{-4}$
Stage 2 Duration	Final 50% of Total Steps (Joint LoRA SFT)
Learning Rates (Stage 2)	$\text{LR}_{\text{bridge}} = \text{LR}_{\text{cross}} = 5\times10^{-5}$, $\text{LR}_{\text{lora}} = 2\times10^{-5}$
Stage 2 Task Replay Mix	80% Patch Synthesis : 20% Comprehension
Partitioned Weight Decay λ_{wd}	10^{-4} on 2D Bridge Matrices; 0.0 on Norms/LoRA

context retrieval alignment.

- **Causal Baseline Curriculum**: Because causal baselines (`Baseline-E2B-IT`, `Baseline-E4B-IT`) have no randomly initialized bridge or cross-attention modules to warm up, they are trained under a continuous **Single-Stage Full Decoder LoRA SFT** across 100% of training steps on the exact same tokenized dataset and step budget.
- **Partitioned Weight Decay**: Weight decay ($\lambda_{\text{wd}} = 10^{-4}$) is applied strictly to 2D linear bridge matrices ($W_{\text{gate}}, W_{\text{up}}, W_{\text{down}}, W_{\text{res}}$) and set to 0.0 on all RMSNorm weights, calibration vectors γ_{calib}, cross-attention gates/scales, and LoRA adapters.

3 Experimental Setup

3.1 Controlled 6-Model Ablation Matrix and Benchmarks

Table 4 details our controlled 6-model experimental matrix spanning symmetric and asymmetric pairings of `Gemma 4 E2B` and `Gemma 4 E4B`. Models are trained on a deterministic multi-file repository corpus combining `SWE-smith` (unified diff bug repair), `LongCodeArena` (`SWE-QA` repository comprehension) [2], and `RepoBench-C-8k` [7]. We evaluate all models across four complementary dimensions:

1. **Long-Context Repository Comprehension (`SWE-QA`)**: Multi-hop architectural reasoning across contexts bucketed at 8K, 16K, and 32K tokens.
2. **Cross-File Code Retrieval (`RepoBench-C-8k`)**: Exact cross-file dependency resolution and completion accuracy.
3. **Automated Bug Localization & Patch Synthesis (`SWE-bench Lite`)**: Unified Diff Validity Rate, File/Hunk Localization Accuracy, and Exact Patch Resolution Rate [6].
4. **Hardware Efficiency Telemetry**: Peak inference VRAM (GB), prefill latency (ms), and decoding throughput (tokens/sec).

3.2 Evaluation Integrity Safeguards

Our benchmark harness enforces two strict invariants: (1) a **Neutral Prompt Contract** providing identical task instructions and the full 32K repository context to both causal baselines (concatenated into the causal prefix up to 40,960 tokens) and BifocalGemma (routed to the 32K bidirectional encoder), without architecture-specific prompt hints, and (2) a **256-Token Comprehension Generation Floor** (`max_new_tokens` ≥ 256) on all MCQ, QA, and retrieval tasks so analytical reasoning preambles are never truncated before emitting the final answer.

4 Empirical Results

[PLACEHOLDER — ACTIVE EXPERIMENTAL RUNS]: *All empirical metrics from earlier exploratory runs have been omitted. This section and Tables 5, 6, and 7 will be populated with final verified metrics upon completion of the active training and benchmark evaluation runs.*

4.1 Long-Context Repository Comprehension

Table 5 reports accuracy across `SWE-QA` (8K, 16K, 32K) and `RepoBench-C-8k`.

- **[PLACEHOLDER]**: Comparative analysis of repository comprehension accuracy (`SWE-QA` and `RepoBench-C-8k`) between Causal Baselines and BifocalGemma across 8K, 16K, and 32K token context buckets.
- **[PLACEHOLDER]**: Analysis of encoder-to-decoder scaling trade-offs (asymmetric `E2B`→`E4B` vs. `E4B`→`E2B`) on cross-file dependency retention.

4.2 Repository-Level Code Repair

Table 6 reports unified diff validity, localization accuracy, and patch resolution on `SWE-bench Lite`.

- **[PLACEHOLDER]**: Evaluation of unified diff synthesis quality, git patch syntactic validity, and file/hunk localization accuracy under the Two-Stage Paired Curriculum and Chunked Differential Diff Cross-Entropy ($\lambda_{\text{ptr}} = 0.0$).

4.3 Hardware Telemetry and Training Dynamics

Table 7 summarizes peak VRAM, prefill latency, and decoding throughput at 32K context length.

- **[PLACEHOLDER]**: Analysis of KV-cache memory efficiency from reusing Cross-KV states across Gemma 4 KV-shared layers and decoding throughput (tok/s) relative to causal baselines.
- **[PLACEHOLDER]**: Summary of Stage 1 (Bridge Alignment) and Stage 2 (Joint LoRA SFT) training and validation loss trajectories, alongside final learned distributions of $\tanh(g_{\text{head}})$ and sharpness σ_{cross}.

5 Limitations and Negative Findings

1. **Dual-Backbone Static Parameter Footprint**: Because BifocalGemma instantiates both a bidirectional encoder and a causal decoder, its static

Table 4: Controlled 6-Model Ablation Matrix for BifocalGemma. All models use native Gemma 4 backbones with stream-isolated Per-Layer Embeddings ($d_{\text{ple}} = 256$) and preserve cross-layer KV-sharing (15/35 KV-owner layers in `E2B`; 24/42 KV-owner layers in `E4B`). Causal baselines use a single-stage full SFT curriculum, whereas BifocalGemma variants use the two-stage paired curriculum (Frozen Decoder Bridge Alignment → Joint LoRA SFT).

Model ID	Topology	Encoder Backbone	Decoder Backbone	Bridge ($d_{\text{enc}} \to d_{\text{mid}} \to d_{\text{dec}}$)	Context ($N_{\text{enc}}/N_{\text{dec}}$)	Curriculum
`Baseline-E2B-IT`	Causal Only	None	`Gemma-4-E2B-IT` (35L)	—	0 / 40,960 (32K+8K)	1-Stage SFT
`Baseline-E4B-IT`	Causal Only	None	`Gemma-4-E4B-IT` (42L)	—	0 / 40,960 (32K+8K)	1-Stage SFT
`Bifocal-2B`	Symmetric	`Gemma-4-E2B` (35L)	`Gemma-4-E2B-IT` (35L)	1536 → 1536 → 1536 (Identity W_{res})	32,768 / 8,192	2-Stage Paired
`Bifocal-4B-E2B`	Dec-Heavy	`Gemma-4-E2B` (35L)	`Gemma-4-E4B-IT` (42L)	1536 → 2560 → 2560 (Ortho W_{res})	32,768 / 8,192	2-Stage Paired
`Bifocal-2B-E4B`	Enc-Heavy	`Gemma-4-E4B` (42L)	`Gemma-4-E2B-IT` (35L)	2560 → 2560 → 1536 (Ortho W_{res})	32,768 / 8,192	2-Stage Paired
`Bifocal-4B`	Symmetric	`Gemma-4-E4B` (42L)	`Gemma-4-E4B-IT` (42L)	2560 → 2560 → 2560 (Identity W_{res})	32,768 / 8,192	2-Stage Paired

Table 5: Long-Context Repository Comprehension and Cross-File Retrieval Results on `LongCodeArena` (`SWE-QA`) across context length buckets (8K, 16K, 32K) and `RepoBench-C-8k` under the Neutral Prompt Contract and 256-token generation floor. *Note: Placeholder cells (`[TBD]`) will be populated upon completion of the active training and evaluation runs.*

Model	`SWE-QA` @ 8K (%)	`SWE-QA` @ 16K (%)	`SWE-QA` @ 32K (%)	`SWE-QA` Overall (%)	`RepoBench-C-8k` (%)
`Baseline-E2B-IT`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`
`Baseline-E4B-IT`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`
`Bifocal-2B` (E2B → E2B)	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`
`Bifocal-4B-E2B` (E2B → E4B)	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`
`Bifocal-2B-E4B` (E4B → E2B)	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`
`Bifocal-4B` (E4B → E4B)	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`

Table 6: Automated Bug Localization and Unified Diff Patch Synthesis on `SWE-bench Lite`. BifocalGemma variants are trained with Chunked Differential Diff Cross-Entropy ($\mathcal{L}_{\text{diff}}$, $\lambda_{\text{ptr}} = 0.0$) and a Two-Stage Paired Curriculum with 20% comprehension replay. *Note: Placeholder cells (`[TBD]`) will be populated upon completion of the active runs.*

Model	Valid Unified Diff Format (%)	File / Hunk Localization Acc. (%)	Exact Patch Resolution (%)	Validation Loss ($\mathcal{L}_{\text{diff}}$)
`Baseline-E2B-IT`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`
`Baseline-E4B-IT`	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`
`Bifocal-2B` (E2B → E2B)	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`
`Bifocal-4B-E2B` (E2B → E4B)	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`
`Bifocal-2B-E4B` (E4B → E2B)	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`
`Bifocal-4B` (E4B → E4B)	`[TBD]`	`[TBD]`	`[TBD]`	`[TBD]`

weight footprint in VRAM is roughly 2× that of a single-stream causal model of identical layer width (`Bifocal-2B` holds ∼ 4B total parameters across encoder and decoder), even though autoregressive decoding steps only traverse the decoder's causal KV cache and the precomputed 15/24 KV-owner cross-attention states.

2. **Sensitivity to Single-Stage Joint Warmup**: Skipping Stage 1 (Frozen Decoder Bridge Alignment) and training decoder LoRA adapters jointly from step 0 alongside unaligned cross-attention projections degrades unified diff formatting. Freezing the decoder during Stage 1 and initializing $\tanh(g_{\text{head}}^{(0)}) = 0.05$ is essential for stable convergence.

3. **Pointer-Generator Copy Bias on Code Repair**: Incorporating an explicit differentiable pointer-generator mixture ($\lambda_{\text{ptr}} > 0$) encourages the decoder to copy contiguous unchanged lines from the encoder context rather than emitting novel bug-fix tokens. Setting $\lambda_{\text{ptr}} = 0.0$ and relying on Chunked Differential Diff Cross-Entropy ($\mathcal{L}_{\text{diff}}$) resolves this failure mode.

4. **[PLACEHOLDER]**: Additional negative findings or boundary conditions observed in the final evaluation runs.

6 Conclusion

We presented **BifocalGemma**, an asymmetric encoder-decoder developer agent architecture built natively on the **Gemma 4** (`E2B` and `E4B`) model family. By combining unmasked bidirectional repository encoding ($N_{\text{enc}} \leq 32{,}768$), a norm-calibrated **SwiGLU Dual-Gate Coordinate Bridge**, **Decoupled Stationary Full-Rank Cross-Attention** ($\Theta_{\text{cross}} = 0$) with **Critical Attention Length Scaling** $\beta(N_{\text{enc}})$ and native Gemma 4 cross-layer KV-sharing, and a **Two-Stage Paired Curriculum** driven by **Chunked Differential Diff Cross-Entropy**, BifocalGemma resolves the structural tension between global repository comprehension and local au-

Table 7: Hardware Telemetry and Inference Efficiency at 32K Repository Context. Only KV-owner layers instantiate dedicated Cross-KV projections, reusing cached $(K_{\text{cross}}, V_{\text{cross}})$ across KV-shared layers during autoregressive generation. *Note: Placeholder cells ([TBD]) will be populated upon completion of the active runs.*

Model	Active Params (Enc + Dec)	KV-Owner Cross-Attn Layers	Peak VRAM @ 32K (GB)	Prefill Latency @ 32K (ms)	Decode Throughput (tok/s)
Baseline-E2B-IT	2.0B	0	[TBD]	[TBD]	[TBD]
Baseline-E4B-IT	4.0B	0	[TBD]	[TBD]	[TBD]
Bifocal-2B	2.0B + 2.0B	15/35	[TBD]	[TBD]	[TBD]
Bifocal-4B-E2B	2.0B + 4.0B	24/42	[TBD]	[TBD]	[TBD]
Bifocal-2B-E4B	4.0B + 2.0B	15/35	[TBD]	[TBD]	[TBD]
Bifocal-4B	4.0B + 4.0B	24/42	[TBD]	[TBD]	[TBD]

toregressive patch generation.

[PLACEHOLDER — FINAL CONCLUSION SUMMARY]: *Empirical conclusions and practical deployment recommendations across symmetric and asymmetric Gemma 4 configurations will be inserted here upon completion of the active training and evaluation runs.*

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